Supporting Information

Analysis Choices and Ranking Stability in Kinase Inhibitor Selectivity Scores

Polina Vinogradova

This document gives the full evidence for results summarized in the main text, in the order of the main text: dataset construction (Section S1), score implementations and agreement (S2), ranking sensitivity and profile associations (S3), the criteria D2 to D4, the gated entropy and its characterization, with the proofs of Propositions 1–3 (S4), recovery from subpanels (S5), cross-dataset and supplementary comparisons (S6), the clinical analysis (S7), and the script for every result (S8). All analyses use the curated panels of the main text, one reference target per human protein kinase, except where Section S1 compares them with the released panels. The scripts in the code repository (`https://github.com/polinavino/kinase-selectivity-definitions`) reproduce every number, and the script for each table and paragraph is named next to it.

S1: Dataset construction and target selection

Released and curated panels. Table S1 compares the main Davis and Klaeger results on the released and curated panels. The released Davis panel lists 433 constructs, 12 of them for EGFR. For 26 of the 68 compounds the two best-bound constructs are forms of one gene. The most ratio-unstable compound on the released panel, GDC-0879, has a top_1–top_2 gap of 0.023, between BRAF(V600E) and BRAF, and a gap of 1.54 to the next gene (RAF1). On Davis, the best affinity per gene is also given as an alternative to the reference construct.

Klaeger et al.[1] annotate 225 of the 343 released Klaeger targets as protein or lipid kinases. The curated panel keeps 217 of them and leaves out three lipid-kinase columns (PIP4K2A, PIP4K2C and PIP4K2A;PIP4K2B), four protein groups (CSNK2A1;CSNK2A3, PDPK1;PDPK2P, PRKCE; PRKCH and PRKX;PRKY), and Q6ZSR9, which UniProt describes as an uncharacterized protein with no kinase domain, encoded in the 3'-UTR of AAK1. The other 118 targets are not annotated as kinases: 46 subunits or adaptors of kinase complexes (for example the cyclins CCNA2–CCNT2, CKS1B, the AMPK and PKA regulatory subunits, and the AP2 adaptor subunits), 7 small-molecule kinases, 20 other enzymes (for example COMT, NQO2, TOP2B, PYGB, PYGL and PTPN1), 43 other proteins, including ribosomal proteins and SLC transporters, and BCR and TRRAP. The source flags 88 of the 118 as possibly co-enriched as complex members, among them BCR (with ABL1) and TRRAP (with CDK8). The 122 targets outside the curated panel, other than the four protein groups, hold 556 of the 2,763 active entries (20.1%), in 122 of the 222 compounds. Six compounds are active only on them (AZD-2014, AZD-8186, NMS-1286937, SGX-523, TG-100115 and volitinib). The code repository lists every target of each dataset with the reason it is kept or removed. Most results change little between the panels. The largest changes are on Davis, where the ratio agrees more with the other definitions on the curated panel, and on Klaeger, where, over all compounds, the Gini correlates more with the ratio than with the S-score on the curated panel.

Klaeger confidence filter. The high-confidence filter removes 793 of the 5,916 drug–target rows (13.4%), one of which has no numeric K_d^{app}. A removed pair is coded at the floor by convention. A

Table S1: Main Davis and Klaeger results on the released and curated panels. Correlations are Spearman r. The "active" rows re-rank the compounds with an active kinase among themselves, as do the "Feature" rows, which correlate a profile feature with instability (Section S3). p^* is from the panel-size analysis without D1 (Section S5). (`davis_gene_level.py`, `klaeger_protein_kinases.py`)

Davis	Released (433)	Curated (363)	Best per gene (363)
Same-gene top two constructs	26	0	0
Table 3: S–E, S–G, E–G	0.946, 0.952, 0.999	0.963, 0.963, 0.999	0.960, 0.960, 1.000
Table 3: R against S, E, G	0.456, 0.343, 0.343	0.563, 0.513, 0.510	0.511, 0.478, 0.474
Feature: n_{active}, entropy	-0.279	-0.301	-0.268
Feature: n_{active}, ratio	$+0.294$	$+0.007$	$+0.189$
Feature: top_1–top_2 gap, ratio	-0.444	-0.445	-0.566
Feature: top_2 pK_d, ratio	$+0.296$	$+0.147$	$+0.203$
Gated entropy vs. entropy	0.956	0.969	0.963
p^* without D1: gated E, E, S, G, R	50, 50, 50, 110, 410	50, 50, 50, 110, 350	50, 50, 50, 110, 350

Klaeger	Released (343)	Curated (217)
Zero-active compounds	16	22
Table 3, all: S–E, S–G, E–G	0.910, 0.722, 0.840	0.896, 0.508, 0.658
Table 3, all: R against S, E, G	0.223, 0.455, 0.616	0.078, 0.344, 0.655
Table 3, active: S–E, S–G, E–G	0.913, 0.911, 0.998	0.897, 0.892, 0.997
Table 3, active: R against S, E, G	0.418, 0.621, 0.634	0.378, 0.622, 0.635
$r(\text{S},\text{G}) - r(\text{G},\text{R})$, active, 95% interval	$[0.195, 0.375]$	$[0.158, 0.355]$
Aggregated rank σ, zero-active vs. active	74.0 vs. 31.2	77.9 vs. 32.2
Feature: n_{active}, entropy	-0.302	-0.312
Feature: top_1–top_2 gap, ratio	-0.291	-0.275
Feature: active range, S-score	-0.315	-0.317
Feature: skewness, Gini	$+0.107$	$+0.072$
Gated entropy vs. entropy	0.983	0.984
p^* without D1: gated E, E, S, G, R	80, 80, 110, 290, > 320	80, 80, 80, 200, > 200

low-confidence fit does not establish weak binding. Of the 792 low-confidence pairs with a numeric K_d^{app}, 405 involve curated targets (375 with a fitted pK_d above 5, and 208 above 6), and 402 of these fall within the 222×217 matrix. Adding them changes 0.77% of its entries (some added values lie below the floor and are set to it). It shifts the all-compound Klaeger correlations of main-text Table 3 by at most 0.08, and those among the 200 compounds with an active kinase under both filters by about 0.006. The relaxed filter makes Brivanib_alaninate and TG-100115 newly eligible, increasing the active population to 202. It leaves every p^* of the main-text recovery analysis unchanged and raises coverage at 50 targets from 70.6% to 72.9%. Retaining all low-confidence pairs also adds 7 compounds and shifts the all-compound correlations by at most 0.06. (`klaeger_filter_sensitivity.py`)

Metz coverage. The 704 retained Metz compounds are measured against 50 to 143 of the 172 kinases (29.1% to 83.1%, with mean 40.9%, median 37.2%, 75th percentile 47.1%). For the median compound, most entries are therefore floor-imputed. Twenty-three compounds have exactly 50 measured kinases and 170 have 55 or fewer. Only 7 kinases are measured for at least 80% of the compounds, so no fully measured subset of useful size exists. (`metz_coverage.py`, `claim_checks.py` section 10)

Activity threshold on Davis. Active entries ($pK_d > 6$) make up 16.7% of the Davis matrix and 4.5% of the Klaeger matrix. The 46 compounds profiled in both are active against 15.0% of the Davis panel and 3.9% of the Klaeger panel on average. With the S-score's threshold sweep shifted up by one log unit on Davis (from $[5.5, 8.0]$ to $[6.5, 9.0]$), the S-score still correlates more with the entropy and Gini (0.88) than with the ratio (0.54). (`dataset_descriptive_stats.py`, `claim_checks.py` section 9)

Datasets not used. The screen of Gao et al.[2] has the same readout type as Anastassiadis, so it would add no new assay technology, and its raw data are not openly redistributable. The study of Patricelli et al.[3] profiles too few inhibitors for a ranking analysis.

S2: Score implementations and agreement

Table S2 gives the correlation of the published, association-constant entropy with the median rankings of the S-score, the log-affinity Gini and the ratio, with floor-coded pairs given zero weight and kept at the floor. It also gives the correlations of the native percent-inhibition Gini on Anastassiadis. (`original_operationalization_robustness.py`)

Table S2: Spearman correlation of the published entropy (association-constant weighting) with the median rankings of the other definitions.

Dataset	Floor-coded pairs	S-score	Gini	Ratio
Davis ($n = 68$)	zero weight	0.738	0.736	0.861
	at the floor	0.344	0.368	0.688
Klaeger ($n = 222$)	zero weight	0.571	0.530	0.657
	at the floor	−0.597	0.077	0.648
Klaeger, active ($n = 200$)	zero weight	0.564	0.782	0.916
	at the floor	−0.412	−0.094	0.591
Metz ($n = 704$)	zero weight	0.291	0.385	0.758
	at the floor	0.241	0.342	0.742

Anastassiadis ($n = 178$): native percent-inhibition Gini (no baseline subtraction) against the median rankings of the other definitions.

	S-score	Entropy	Ratio
Gini (native)	0.340	0.629	0.326

In every case the published entropy correlates more with the ratio than with the S-score or the Gini, so it would not be grouped with them on any of the three affinity datasets. How far it is from the S-score and Gini depends on the treatment of floor-coded pairs. On Klaeger, a third treatment keeps each of the 4,000 measured high-confidence affinities on the curated panel at its measured value, including the 194 weaker than the floor (in 84 compounds), and omits unmeasured pairs. The published entropy then correlates with the S-score, Gini and ratio at 0.558, 0.531 and 0.663 over all compounds and 0.558, 0.778 and 0.918 over the 200 with an active kinase, against 0.571, 0.530, 0.657 and 0.564, 0.782, 0.916 with floor-coded pairs given zero weight. The published entropy's rankings under the two treatments correlate at 0.9987 over all compounds and 0.9996 over active ones. On the subpanels of the recovery analysis, p^* stays at 200 targets, and the mean Spearman correlation at 50 targets is 0.6025 against 0.6014.

Figure S1 shows the correlations between the four definitions on all four datasets, among compounds with an active kinase and among all compounds. Among the S-score, entropy and Gini, the correlations among active compounds are 0.963–0.999 (Davis), 0.892–0.997 (Klaeger), 0.870–0.999 (Anastassiadis) and 0.806–0.994 (Metz), and the ratio's correlations with them 0.510–0.563, 0.378–0.635, 0.793–0.843 and 0.147–0.203. With all compounds, the ranges are 0.508–0.896 and 0.078–0.655 on Klaeger, 0.859–0.988 and 0.478–0.553 on Anastassiadis, and 0.810–0.994 and 0.138–0.193 on Metz. Using mean instead of median ranks changes no all-compound correlation by more than 0.036 (Davis), 0.076 (Klaeger), 0.075 (Anastassiadis) or 0.052 (Metz). (`additional_datasets_analysis.py`, `claim_checks.py` sections 3 and 11)

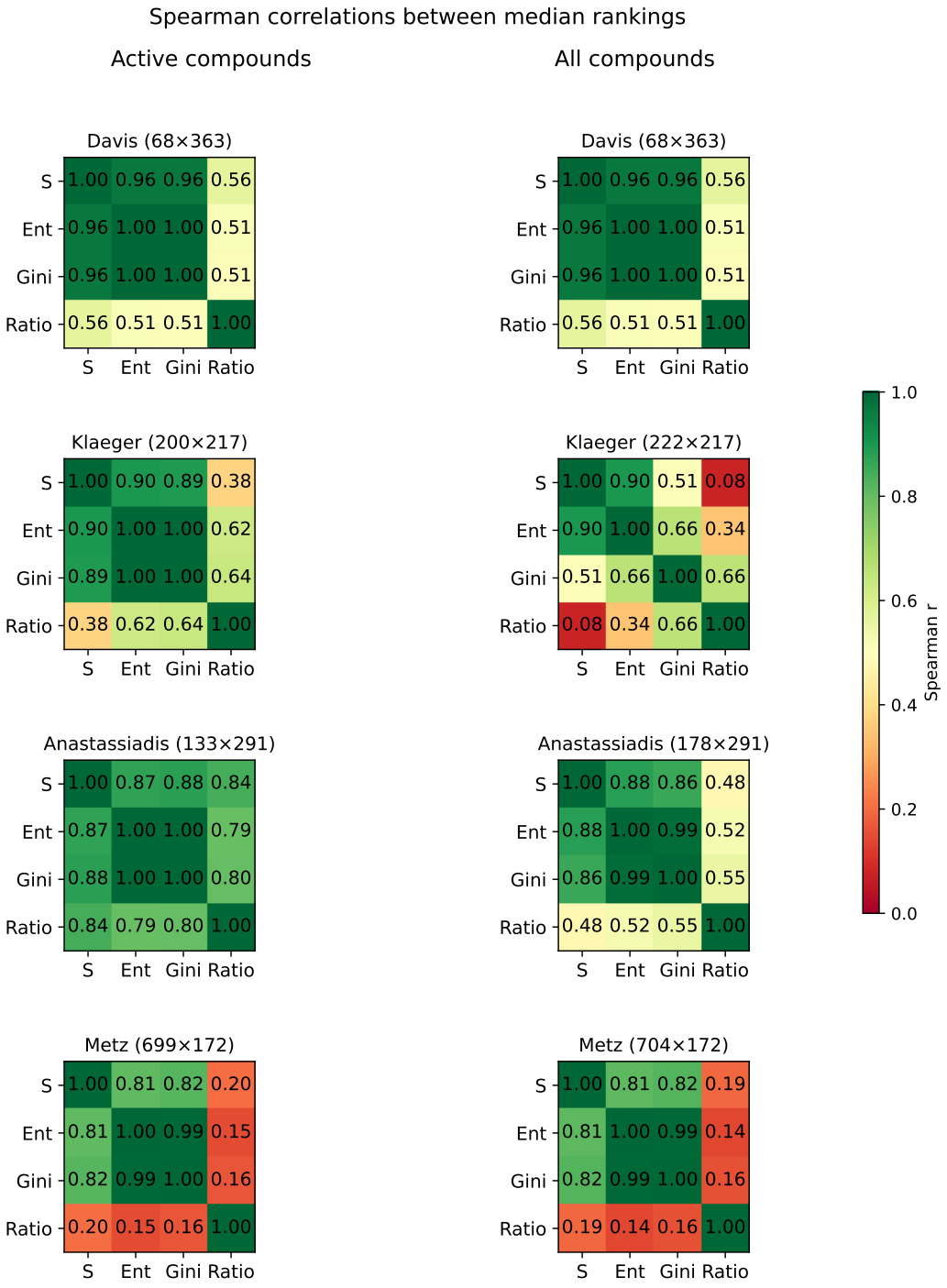


Figure S1: Spearman correlations between the median rankings of the four definitions on each dataset (compounds × targets above each panel). Left: compounds with an active kinase, ranked among themselves. Right: all compounds. S is the S-score and Ent the entropy.

S3: Ranking sensitivity and profile associations

Compounds with no active kinase. Compounds with no active kinase have higher aggregated rank instability than active compounds on every dataset where they occur (77.9 against 32.2 on Klaeger, 36.6 against 21.8 on Anastassiadis, and 178.5 against 108.1 on Metz, which has five such compounds). The contrast comes mainly from the definitions disagreeing. Within a single definition's sweep on Klaeger, it is significant only for the Gini (65.0 against 16.3, $p < 0.001$). It is small and not significant for the entropy (11.9 against 12.6, $p = 0.91$) and the ratio (17.5 against 19.1, $p = 0.51$), and reverses for the S-score (19.0 against 25.7, $p = 0.01$), whose score is 0 for every such compound at thresholds of 6 or above. On Anastassiadis it is significant for the Gini, the entropy and the ratio. On Metz it reverses for the S-score, the entropy and the Gini (7.8 against 71.2, 11.0 against 24.5, and 5.4 against 24.2, all with $p < 0.05$). On Klaeger, the aggregated contrast persists with the undefined entropy and Gini configurations dropped (66.2 against 31.7) and with only the S-score and ratio aggregated (70.6 against 37.7, both $p < 0.001$). With instability computed only over each compound's defined configurations, the 16 Klaeger compounds with no active kinase that have at least two defined configurations have lower mean instability than active ones within the entropy (7.0 against 13.5) and the Gini (6.7 against 13.6, both $p = 0.002$). This is a group result: three of the 16 (brivanib, brivanib alaninate and NMS-1286937) are less stable than the active mean, and the compounds contribute different numbers of configurations. (`claim_checks.py` sections 13 and 20, `undefined_score_convention.py`)

Six Klaeger compounds with no detected binding. AZD-2014, AZD-8055, AZD-8186, BMS-911543, SGX-523 and volitinib show no binding to any curated target, although they are established inhibitors of mTOR, mTOR, PI3Kβ, JAK2, MET and MET. mTOR and PI3Kβ were not quantified for any of the 229 compounds with a fitted target in the Kinobeads table, consistent with low expression or poor capture in the lysate. PI3Kβ, a lipid kinase, is also outside the curated panel. JAK2 and MET were quantified for 7 and 32 other compounds (up to p$K_d = 8.3$ and 9.5), so absence from the lysate does not explain the result for BMS-911543, SGX-523 and volitinib. These data do not show whether compound-specific capture, competition or curve fitting explains it. (`claim_checks.py` section 1)

Boundary sensitivity. Table S3 gives the share of pairs of scored compounds whose order reverses as each parameter moves across $B = [x_0, \tau^*]$. B is $[5.0, 6.0]$ on Davis and Klaeger, $[4.0, 6.0]$ on Metz and $[20, 50]$% inhibition on Anastassiadis. For the ratio, the parameter is the level b in $R_b(k) = x_{(1)} - \max(x_{(k+1)}, b)$. The gated entropy's gate is swept with T fixed at $\tau^* - x_0$. Every parameter reverses more than 13% of pairs on at least one dataset (the ratio's level only at $k = 5$). Figure S2 shows the per-compound rank instability on Klaeger. (`d3_pair_reversals.py`, `failure_mode_illustration.py`)

Profile features and instability. Table S4 gives the correlations between profile features and within-definition instability, with the compounds that have an active kinase ranked among themselves, and, for Klaeger, ranked among all 222 compounds. The ranking population changes the Gini column most: ranked among all compounds, the skewness of the active affinities is the Gini's strongest correlate (+0.262), and ranked among active compounds it is not significant (+0.072), while the number of active kinases becomes significant (−0.254). Over the 189 active compounds whose entropy and Gini are defined throughout the baseline sweep, re-ranked among themselves, the number of active kinases correlates with entropy and Gini instability at −0.274 and −0.272

Table S3: Boundary sensitivity: pairs of scored compounds reversing in B (%), for every swept parameter. The S-score's values are exact. The other parameters are evaluated at 201 equally spaced values, and the gate also at every observed value in B and just below it, where its score jumps. These values are estimates: with 101 values instead of 201, none changes by more than 0.01 percentage point.

Parameter swept	Davis	Klaeger	Anastassiadis	Metz
S-score, threshold τ	8.43	9.96	12.99	21.82
Entropy, baseline β	7.90	12.37	10.58	22.80
Gini, baseline β	8.56	12.40	10.51	22.49
Gated entropy, gate g	9.22	12.66	14.56	23.82
Ratio, level b, $k = 1$	0.00	3.00	2.13	0.04
Ratio, level b, $k = 2$	0.83	6.26	4.10	0.43
Ratio, level b, $k = 3$	1.10	9.19	6.08	0.75
Ratio, level b, $k = 4$	1.71	12.03	8.17	0.84
Ratio, level b, $k = 5$	2.59	13.71	9.72	1.03

(both $p < 0.001$). Figure S3 plots each definition's strongest correlate. With a Bonferroni correction for the 24 tests per dataset, 8 of the 13 nominally significant Klaeger correlations survive, and on Davis only the ratio–gap correlation. The Holm procedure gives the same result. The Benjamini–Hochberg procedure keeps all 13 Klaeger correlations and, on Davis, again only the ratio–gap correlation. On Anastassiadis and Metz, with active compounds ranked among themselves, the number of active kinases correlates with entropy instability at -0.38 and -0.42 (both $p < 0.001$), and the top_1–top_2 gap with ratio instability at 0.00 ($p = 0.97$) and -0.26 ($p < 0.001$). Without the eight Anastassiadis compounds whose two best targets are both at 99.5% inhibition or more, the gap–ratio correlation there is -0.15 ($p = 0.10$). (`instability_by_family.py`, `additional_datasets_analysis.py`, `claim_checks.py` sections 4 and 21)

Undefined features. The skewness of the active affinities is undefined with fewer than three active kinases, and their standard deviation and range with fewer than two. Table S4 sets these to 0. Over only the compounds for which each feature is defined, the Klaeger range associations strengthen (S-score, entropy and Gini -0.373, -0.378 and -0.349, all $p < 0.001$), the Gini–skewness association stays non-significant ($+0.145$, $p = 0.08$), and no Davis correlation changes significance. (`feature_placeholder_sensitivity.py`)

Range and skewness. Range rises with the number of active kinases (Spearman 0.76 on Klaeger). With that number held fixed, range remains associated with S-score instability (partial $r = -0.22$, $p = 0.002$) but not with entropy or Gini instability ($+0.08$ and $+0.01$). On the 46 compounds profiled in both Davis and Klaeger, the range–S-score association is weak on both (Klaeger $r = -0.10$, $p = 0.56$, for the 37 with at least two active kinases, and Davis -0.16, $p = 0.30$). These compounds have a median of 4 active kinases on Klaeger and 32 on Davis. Among the 148 Klaeger compounds with at least three active kinases, skewness correlates with the share of a compound's weight on kinases that the β sweep crosses (0.38), and that share with Gini instability with the number of active kinases held fixed (partial $r = 0.43$). Ranked among active compounds, skewness has no marginal association with Gini instability, but with the number of active kinases held fixed it does (partial $r = 0.35$, $p < 0.001$). Holding the weight share fixed as well lowers this to 0.19 ($p = 0.02$). The share of a compound's above-floor kinases that some baseline crosses correlates

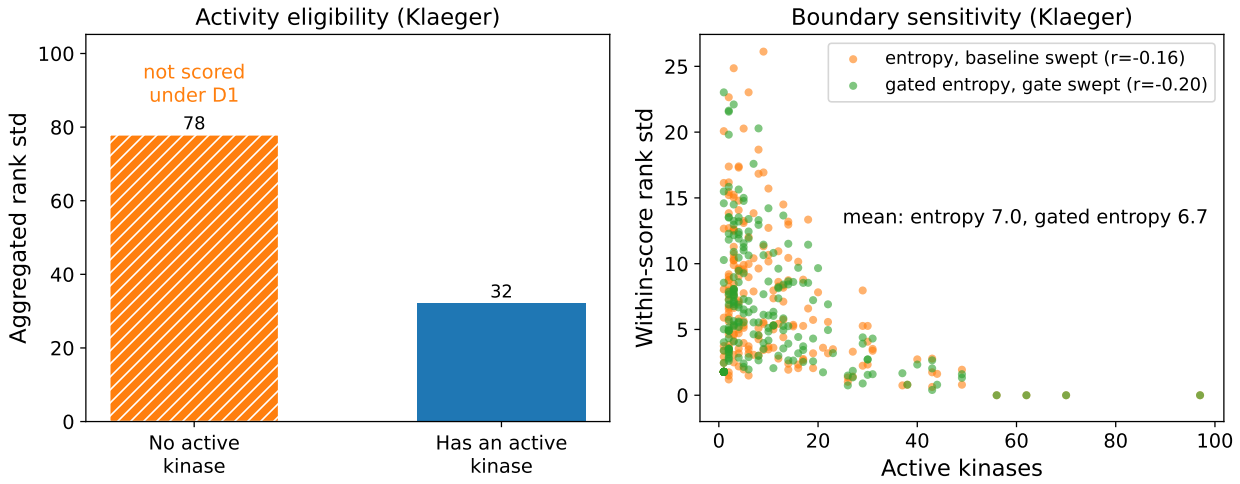


Figure S2: Rank instability on Klaeger. *Left (D1):* rank standard deviation across the four definitions' parameter sweeps, for the compounds with no binding above the activity threshold and for those with an active kinase, all ranked together. *Right:* per-compound rank standard deviation, among compounds with an active kinase ranked among themselves, against the number of active kinases, with each score's boundary swept over $B = [5.0, 6.0]$: the entropy's baseline β and the gated entropy's gate g.

with its entropy instability ($r = 0.22$, $p = 0.002$). (`claim_checks.py` sections 8 and 15–17)

Further features. Five further features were tested on Klaeger and Davis: the top-target concentration ratio CR_1 and the Herfindahl–Hirschman Index (HHI) from industrial-organization economics, the primary-target affinity, a near-tie count generalizing the top_1–top_2 gap, and the active-affinity coefficient of variation. CR_1 and HHI are near-collinear with n_{active} ($r = -0.98$ for CR_1, and -0.997 and -0.998 for HHI). The other three were weaker than the strongest feature of Table S4 for every definition and on both datasets, except the primary-target affinity, which correlates with Gini instability on Klaeger at -0.295. (`instability_predictor_candidates.py`)

Unstable and stable compounds. All 20 most unstable Klaeger compounds, ranked together, have no active kinase. Among the 20 most unstable compounds with an active kinase, ranked among themselves, five have a top_1–top_2 gap below 0.1, five have at least 10 active kinases, and ten have neither. Of the ten largest entropy–ratio disagreements among active Klaeger compounds, seven have 14–27 active kinases, six of them with a top_1–top_2 gap above 1, for example dovitinib ($n_{\mathrm{active}} = 22$, gap 1.93) and BMS-690514 (20, 1.98). The ratio ranks these seven 31st to 63rd of 200, and the entropy and Gini 146th to 180th. The other three have one to three active kinases and are ranked as more selective by the entropy: fasudil (3 active kinases, gap 0.03, ratio rank 181, entropy rank 68), CUDC-101 (3, 0.19, 172, 61) and rigosertib (1, 1.03, 118, 10). The most stable active Klaeger compounds are among the most promiscuous: XL-228 ($\sigma = 3.0$, 49 active kinases) and PF-03814735 (5.4, 40) have median ranks of 190–198 of 200 under every definition, and mean instability falls from 25–32 for compounds with 1 to 40 active kinases to 15 above 40. On Davis, compounds with few active kinases are the most stable on average (mean $\sigma = 3.9$ for one or two, 4.5 for three to five, and 7.3 for more than 40). The most stable, nintedanib ($\sigma = 2.1$), has 153 active targets and median ranks of 59–62 of 68. Without the ratio's configurations, the

Table S4: Spearman correlations between profile features and within-definition rank instability. Top: Klaeger (200 compounds with an active kinase, ranked among themselves) and Davis (68). Bottom: the same 200 Klaeger compounds, ranked among all 222. S is the S-score, E the entropy, G the Gini and R the ratio. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

	Klaeger, active ranking				Davis			
Feature	S	E	G	R	S	E	G	R
n_{active}	-0.236^{***}	-0.312^{***}	-0.254^{***}	-0.040	-0.210	-0.301^{*}	-0.313^{**}	$+0.007$
top_1–top_2 gap	$+0.074$	$+0.019$	$+0.016$	-0.275^{***}	$+0.043$	$+0.074$	$+0.058$	-0.445^{***}
Active affinity std	-0.254^{***}	-0.083	-0.135	-0.129	-0.017	$+0.020$	$+0.035$	-0.176
Active affinity range	-0.317^{***}	-0.185^{**}	-0.188^{**}	-0.118	-0.023	-0.065	-0.076	-0.240^{*}
top_2 pK_d	-0.271^{***}	-0.157^{*}	-0.218^{**}	$+0.188^{**}$	-0.174	-0.186	-0.190	$+0.147$
Active affinity skew	-0.199^{**}	-0.051	$+0.072$	-0.089	$+0.041$	$+0.195$	$+0.176$	-0.113

	Klaeger, ranked among all 222			
Feature	S	E	G	R
n_{active}	-0.253^{***}	-0.394^{***}	-0.075	-0.025
top_1–top_2 gap	$+0.016$	$+0.057$	$+0.057$	-0.342^{***}
Active affinity std	-0.308^{***}	-0.132	-0.056	-0.121
Active affinity range	-0.370^{***}	-0.247^{***}	-0.054	-0.113
top_2 pK_d	-0.300^{***}	-0.230^{**}	-0.219^{**}	$+0.221^{**}$
Active affinity skew	-0.236^{***}	-0.096	$+0.262^{***}$	-0.107

mean instability of the 32 Davis compounds with more than 40 active targets falls from 7.3 to 3.7. (`claim_checks.py` sections 5–7 and 18)

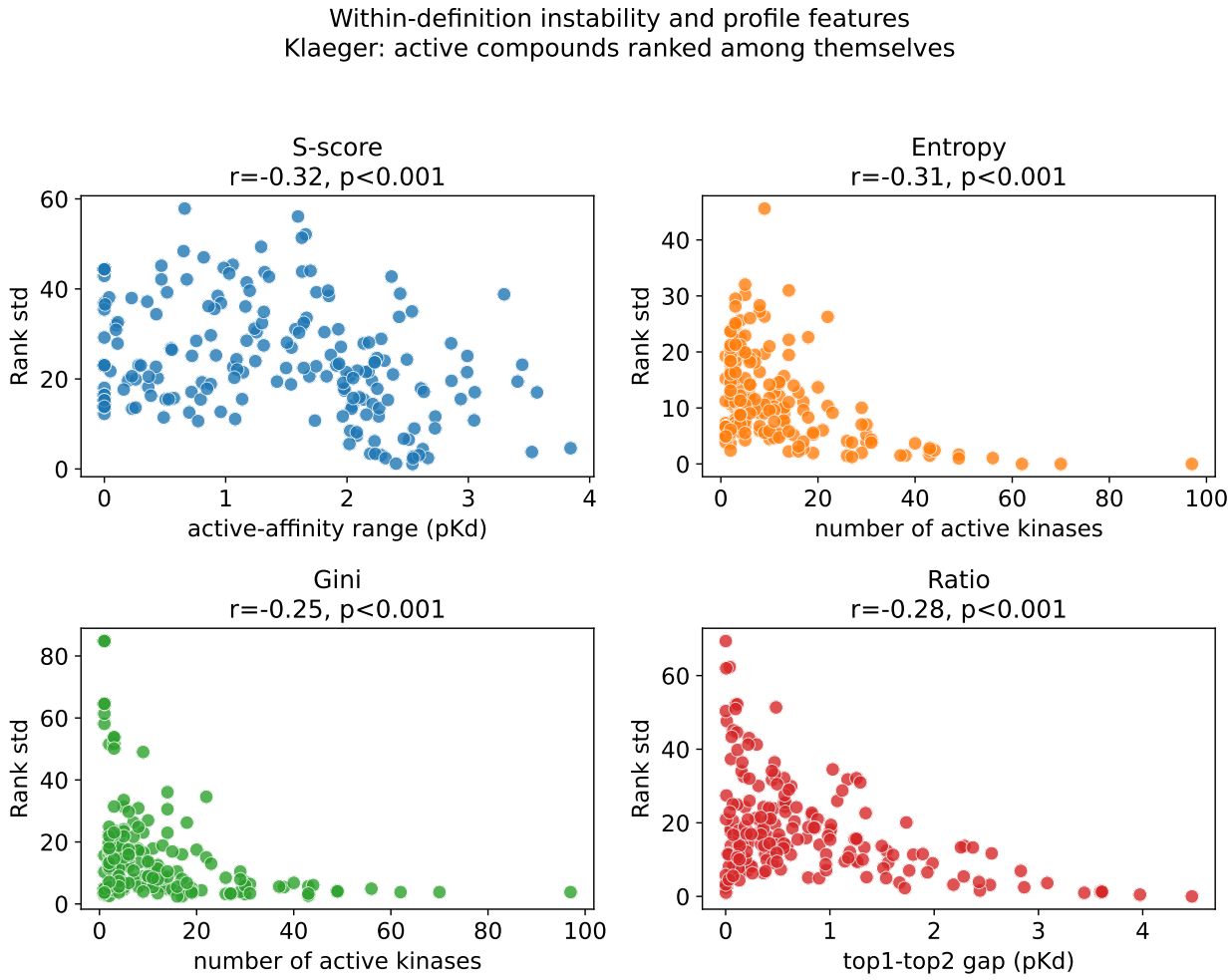


Figure S3: Within-definition rank instability against the profile feature most strongly correlated with it (Klaeger, the 200 compounds with an active kinase, ranked among themselves): active-affinity range (S-score), number of active kinases (entropy and Gini), and top_1–top_2 gap (ratio). Spearman r above each panel.

S4: The criteria D2 to D4, the gated entropy and its characterization

Proof of Proposition 1. With the gate at g, every gated kinase has weight $w_i = e^{(x_i - g)/T} > 1$. A kinase added at or below g has weight 0 and leaves the score unchanged. A kinase added with value in $(g, \tau^*]$ has weight $a = e^{(x-g)/T} \le e^{(\tau^* - g)/T}$, which is at most e when $T \ge \tau^* - g$. Let the original profile have total weight $W = \sum_i w_i$ and $L = \sum_i w_i \ln w_i$, so that its entropy in nats is $H = \ln W - L/W$. After the addition,

$$H' - H = \ln\Big(1 + \frac{a}{W}\Big) + \frac{a}{W + a}\left(\frac{L}{W} - \ln a\right).$$

Every $w_i > 1$, so $L > 0$, and $a \le e$, so $\ln a \le 1$. With $t = a/W > 0$ this gives $H' - H > \ln(1+t) - t/(1+t) > 0$. The entropy therefore rises, so the score does not. □ With $T = \tau^* - x_0$ and $g \in B$, $\tau^* - g \le T$, so the condition holds for every gate in B. A random check over 20,000 weight vectors with every weight at least 1 found no case in which appending a weight of at most e lowers the entropy, and 1,537 cases when the appended weight is $1.05\,e$.

Sharpness (*only if* part). The bound cannot be lowered. If $T < \tau^* - g$, a kinase added at τ^* has weight $r = e^{(\tau^* - g)/T} > e$. Take one active target with weight s and n kinases just above the gate, with weights close to 1. For the limiting weights the entropy is $H_n = \ln(n+s) - s\ln s/(n+s)$, and after the addition $H'_n = \ln(n+s+r) - (s\ln s + r\ln r)/(n+s+r)$. Then $n(H'_n - H_n) \to r(1 - \ln r) < 0$ as $n \to \infty$, so for large enough n the entropy falls and the score rises. By continuity, this also holds for weights slightly above 1, as the strict gate requires. Numerically, with T 3% below $\tau^* - x_0$, a profile with one target just above τ^* and 280 kinases just above the floor violates D3 by 2.2×10^{-4} bits on Klaeger, Metz and Anastassiadis, and with T 1% above the bound no such profile does. A random search over 20,000 profiles per dataset found no violation with $T = \tau^* - x_0$, and adding one kinase at τ^* never raises the score of a real compound. (`d4_entropy_vs_candidate.py` sections 1–5 and 14)

Proof of Proposition 2. *D4 and grouping fix the aggregate.* This step does not use h. On the profiles $\mathbf{v}_m$, the positive weights are m equal weights, so D4 requires $A_F(m) = F(1/m, \ldots, 1/m)$ to be nondecreasing in m and $A_F(N) > A_F(1)$. Shannon[4] (Theorem 2) showed that a continuous F with the grouping property and increasing A_F is $F(\mathbf{p}) = -K\sum_i p_i \log p_i$. His argument uses the monotonicity of A_F only to show that $A_F(n) = K \log n$, for which nondecreasing A_F suffices, and $A_F(N) > A_F(1)$ gives $K > 0$. *D2 fixes the weights.* Take a profile with two targets above τ^*, at distances $u \ge v > 0$ above the gate placed at τ^*, and lower the gate by d with $0 < d \le \tau^* - x_0$. The gate passes no value, so D2 requires the score, and therefore the Shannon entropy of the two shares, to be unchanged. For two shares, this entropy falls strictly as the ratio of the larger weight to the smaller rises from 1, and h is nondecreasing, so the ratio is unchanged: $h(u+d)/h(v+d) = h(u)/h(v)$. Hence $A(d) = h(u+d)/h(u)$ does not depend on u. Then $A(d_1 + d_2) = A(d_1)\,A(d_2)$ whenever $d_1 + d_2 \le \tau^* - x_0$, and A is continuous, so $A(d) = e^{cd}$ with $c \ge 0$, because h is nondecreasing. Repeated shifts give $h(t) = Ce^{ct}$ for every $t > 0$. With $c = 1/T$ these are the gated entropy's weights, and $c = 0$ gives equal weights. The score is therefore K times the gated entropy with this T. The Rényi entropy of order $\alpha \ne 1$ is continuous and increases with n on equal shares but lacks the grouping property. □

Proof of Proposition 3. Let $W = \sum_i w_i$ and sort the original positive weights in decreasing order. Because $a \le \min_i w_i$, the added weight comes last among the positive weights. For $j \le r$ the sum of the j largest new shares, $\sum_{i \le j} w_{(i)}/(W + a)$, is at most the corresponding old sum, $\sum_{i \le j} w_{(i)}/W$, and for $j > r$ both sums equal 1. The new shares are therefore more even than the old ones (the old shares majorize the new ones), so f does not increase. □ Functions f that never increase when the shares become more even are called Schur-convex. The negative Shannon and Rényi entropies have it because the entropies are Schur-concave. The largest share has it, and here falls to $w_{(1)}/(W + a)$. The Gini coefficient has it on a fixed panel. For the Shannon entropy, the formula for $H' - H$ in the proof of Proposition 1 also gives the result directly: $L/W \ge \ln a$ because every $w_i \ge a$, so the first term is positive and the second non-negative. The condition on a is needed: the weights 10 and fifty 1s have $H = 5.353$ bits, and adding another 10, tied with the strongest, lowers it to 5.180 bits. Without the condition, a concentration score can also rank a profile with more active targets as more selective, as it should when binding is more concentrated: $H(1, 1) = 1$ bit, but $H(1000, 1, 1) = 0.023$ bits.

Checks of D4 and Proposition 3. On reference profiles $\mathbf{v}_m$ with $a = 6.5$, 7, 8 and 9 on a 30-kinase panel, every score in Table 4 of the main text, $P_{\max}$ and the published entropy with floor-coded pairs given zero weight satisfy D4. The active count with its sign reversed and the gated entropy without its non-binder gate fail it. A random check of Proposition 3 over 20,000 weight vectors found no violation. Adding one kinase at $\mathrm{p}K_d = 12$ to a profile with one target at 9 and fifty at 5.1 raises the gated entropy's score from -3.839 to -0.790, the entropy's from -4.127 to -3.310 and $P_{\max}$ from 0.994 to 0.999, and lowers the active count's from -1 to -2: adding an active target is not constrained by D4. Such an addition need not raise a concentration score either: adding a weight of 2 to a profile whose only positive weight is 1 lowers $-H$ from 0 to -0.918 bits. D4 also depends on the floor treatment: if floor-coded pairs keep weight 1 and a target has weight 2, the change from $(2, 1, 1, 1)$ to $(2, 2, 1, 1)$ raises $-H$ from -1.922 to -1.918 bits, so a broader profile is not then a more even one. The published entropy with floor-coded pairs kept at the floor fails D4 in this way: with targets at $\mathrm{p}K_d = 6.01$ on every panel size checked (30 to 363 kinases), at 7 on 172 kinases or more, and at 8 on 363, while it holds at 9. On 363 kinases with targets at 7, a second target raises its score from -7.414 to -6.763 bits. $P_{\max}$ satisfies D4 under either floor treatment. A baseline below the floor also gives floor-coded pairs positive weight. Over the Anastassiadis sweep (291 kinases, floor 20%, targets at 55% to 100%), the entropy and Gini satisfy D4 for $\beta \ge 20\%$ and fail it at every tested target value for $\beta = 0$, 5, 10 and 15%. With $\beta = 0\%$ and targets at 60%, a second target raises the Gini from 0.00680 to 0.01347 and the entropy's score from -8.1785 to -8.1723 bits. (`d4_entropy_vs_candidate.py` sections 9–12)

Check of D2. Over 5,000 random profiles with no value in $[5.0, 6.5]$, moving the gate from 5.0 to 6.5 leaves the gated entropy unchanged. The score changes for 85.9% of the profiles when each weight is instead $\ln(1 + e^d)$, where d is the target's distance above the gate, and for 85.9% under the entropy with its baseline as the boundary. For two targets 1.0 and 0.5 units above the gate, lowering the gate by 0.5 leaves the ratio of the exponential weights at 1.649 and changes the ratio of the weights $\ln(1 + e^d)$ from 1.348 to 1.296. (`d4_entropy_vs_candidate.py` section 13)

Validation. Table S5 gives the checks of the gated entropy on each dataset. Removing the non-binder gate reverses its correlation with the number of active kinases (gated $r = -0.935$, ungated $r = +0.67$ on Klaeger), and the ungated score fails D4. (`candidate_validation.py`)

Table S5: The gated entropy: correlation with the number of active kinases (r with n_{active}), pairs reversing in B (%) when the gate or the entropy's baseline is swept, and D3, the percentage of compounds whose score does not rise when one kinase is added at τ^*, over the n compounds that D1 allows to be scored.

Dataset	n	r with n_{active}	Gate swept	Entropy's baseline swept	D3 (%)
Davis	68	−0.947	9.22	7.90	100.0
Klaeger	200	−0.935	12.66	12.37	100.0
Anastassiadis	133	−0.926	14.56	10.58	100.0
Metz	699	−0.761	23.82	22.80	100.0

Variants. D3 is proved for the Shannon order with $T \geq \tau^* - x_0$. For the other variants it is checked by adding one kinase at τ^* to every scored compound, and it held for all of them. *Quantile floor.* Using a per-compound low quantile of the profile (the 5th to 15th percentile) as the gate gives a variant invariant to shifts of the whole profile, but on these data it coincides with the gated entropy. On Davis every such percentile equals the floor for all but at most two compounds, and on Klaeger for every compound. At the 10th percentile the variant ranks compounds as the gated entropy does (Spearman 1.000), and across the five percentiles no pair reverses. *Rényi order.* Replacing the Shannon entropy by the Rényi entropy of order α, $-H_\alpha(p) = \frac{1}{\alpha-1}\log_2 \sum_i p_i^\alpha$, gives a score that still satisfies D1 and D2. Orders 0.5 to 3 rank compounds as the Shannon order does at Spearman 0.93 or higher (Table S6). Without D1 (Section S5), orders from 0.01 to 1 reach $p^* = 80$ on Klaeger and Metz, and orders 2 and 3 reach it at 110 on Klaeger and 80 on Metz. *Scale.* On Klaeger, over $T = 0.5$, 0.75, 1, 1.5 and 2, the correlation with the number of active kinases ranges from −0.776 ($T = 0.5$) to −0.945, and D3 held for every scored compound at every scale, including the two below the bound of Proposition 1. Across these scales the rankings agree at Spearman 0.79 or higher, and for $T \geq 1$ at 0.985 or higher. *Other weights.* Equal weights (the limit $T \to \infty$) and the association-constant weights of the published entropy ($T = 1/\ln 10$) rank compounds with Spearman 0.963–0.978 and 0.773–0.825 against the gated entropy on Davis and Klaeger (Table S6). (`family_construction.py`, `family_panelsize.py`, `candidate_validation.py`)

What the criteria determine. Proposition 2 characterizes the scores in one class: each target above the gate is weighted by the same function of its distance above the gate, and the score is an aggregate of the weight shares with Shannon's grouping property. Within this class, D2 and D4 leave the gated entropies, with any $T > 0$, and the equal-weight limit, and D3 leaves those with $T \geq \tau^* - x_0$ (Proposition 1). The class assumption is needed. If each weight is a function of the distance above a fixed reference and only the gate moves, D2 holds for every weight function and does not determine the weights. The criteria also leave T free above $\tau^* - x_0$. We choose the smallest admissible value, $T = \tau^* - x_0$, which makes the score depend most on affinity differences. The published entropy, with floor-coded pairs given zero weight, is the member with $T = 1/\ln 10 \approx 0.43$ pK units, below the bound on all three pK datasets, which is why it violates D3.

Noise. With random noise added to the measured values, as described in the main text (Methods), the number of compounds whose D1 status changed in a draw was at most 4 (Klaeger, noise 0.1) and 11 (Klaeger, 0.3), at most 10 on Anastassiadis (9 percentage points) and 5 on Metz (0.3), and none on Davis. Over the compounds scored with and without noise, the gated entropy's ranking correlated with the noise-free ranking at Spearman 0.972 or higher (lowest on Anastassiadis), and the entropy's at 0.983 or higher. (`claim_checks.py` section 23)

Table S6: Variants of the gated entropy on Davis and Klaeger, over the compounds D1 scores: Spearman correlation with the gated entropy's ranking. D3 held for every scored compound under every variant. (`family_construction.py`)

Variant	Davis	Klaeger
Rényi order $\alpha = 0.5$	0.984	0.992
Rényi order $\alpha = 2$	0.968	0.977
Rényi order $\alpha = 3$	0.932	0.953
Equal weights ($T \to \infty$)	0.963	0.978
Association constants ($T = 1/\ln 10$)	0.773	0.825
Quantile floor, 10th percentile	1.000	1.000

S5: Recovery from subpanels

Without D1. Karaman et al.[5] scored each compound's S-score on random subpanels and found that it varies more as panels shrink. Repeating that design for the four definitions, scoring every compound on every subpanel, makes the ranking depend also on each definition's convention for compounds that have no active target on the subpanel. For each panel size, in steps of 30 from 50 targets up to the full panel, we drew 50 random subsets, computed each definition on the subpanel with the dataset's floor x_0 as its baseline (entropy, Gini) or level (ratio, $k = 1$) and the activity threshold as the S-score cutoff, and correlated its ranking with the full-panel ranking. The S-score recovers the full-panel ranking fastest on every dataset, and the entropy as fast on Davis, Klaeger and Anastassiadis (Figure S4). Their p^* is 50 targets on Davis and Anastassiadis and 80 on Klaeger, and on Metz 50 for the S-score and 80 for the entropy. The ratio is the slowest, needing 350 of 363 targets on Davis, 260 of 291 on Anastassiadis, 170 of 172 on Metz, and more than the largest subpanel tested on Klaeger ($r = 0.887$ at 200 targets). The Gini needs 110 targets on Davis and 200 on Anastassiadis and Klaeger, but matches the entropy on Metz (80). Its slower recovery comes from compounds whose Gini is undefined on the subpanel, which the convention ranks least selective. Excluding them gives the Gini p^* values of 50, 80, 50 and 80 targets. Changing the bar within $[0.80, 0.95]$ changes the p^* values but never reverses the order of two definitions on any dataset (Table S7). (`panel_size_analysis.py`, `pstar_bar_sensitivity.py`, `claim_checks.py` section 14)

Table S7: p^* (targets) without D1 at four correlation bars. (`pstar_bar_sensitivity.py`)

Davis (363 targets)	entropy	Gini	S-score	ratio
bar = 0.80	50	80	50	320
bar = 0.85	50	110	50	320
bar = 0.90	50	110	50	350
bar = 0.95	50	170	80	350

Klaeger (217 targets)	entropy	Gini	S-score	ratio
bar = 0.80	50	170	50	200
bar = 0.85	50	200	80	200
bar = 0.90	80	200	80	> 200
bar = 0.95	110	200	140	> 200

Anastassiadis (291 targets)	entropy	Gini	S-score	ratio
bar = 0.80	50	170	50	260
bar = 0.85	50	170	50	260
bar = 0.90	50	200	50	260
bar = 0.95	80	260	80	290

Metz (172 targets)	entropy	Gini	S-score	ratio
bar = 0.80	50	50	50	170
bar = 0.85	50	50	50	170
bar = 0.90	80	80	50	170
bar = 0.95	110	110	80	170

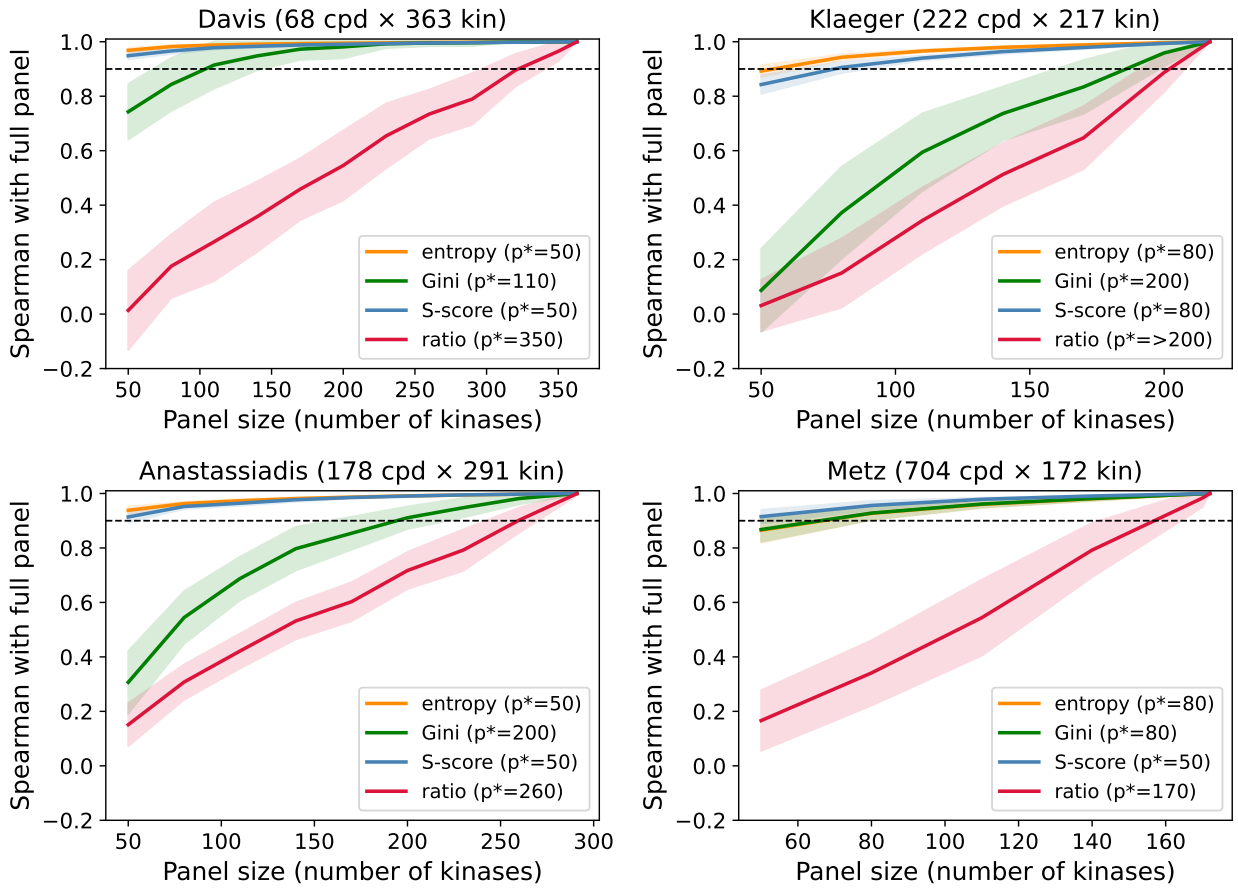


Figure S4: Mean Spearman correlation between rankings computed on random target subpanels and rankings on the full panel, as a function of subpanel size, with every compound scored on every subpanel. Shaded bands: ±1 standard deviation across 50 subpanels. Dashed line: $r = 0.90$. The value p^* (legend) is the smallest panel reaching mean $r > 0.90$.

Scale. With $T = 2(\tau^* - x_0)$ and $4(\tau^* - x_0)$, the gated entropy reaches p^* at 20, 50, 30 and 65 targets on Davis, Klaeger, Anastassiadis and Metz, as the share-following scores do. Its full-panel rankings correlate with those of the S-score with $\tau = x_0$ at 0.9968–0.9996 and 0.9994–0.9998. Both scales satisfy D1 to D4 (Proposition 1). (`panel_recovery.py`)

Variability across subpanels. In the main-text recovery analysis, with D1 applied, p^* is a threshold on the mean over 50 subpanels. At p^*, the share of individual subpanels above 0.90 is 64%, 68%, 78% and 88% for the gated entropy on Davis, Klaeger, Anastassiadis and Metz, 64–90% for the S-score at x_0, 66–82% for the entropy at x_0, 62–74% for the Gini, 60–80% for the S-score at τ^*, and 78–90% for the entropy at τ^*. Keeping floor-coded pairs at the floor instead of giving them zero weight leaves the published entropy's p^* unchanged (300, 200 and 150 targets). With bars of 0.80, 0.85 or 0.95 instead of 0.90, the share-following scores (the S-score and entropy at x_0 and the Gini) reach the bar no later than the S-score and the entropy at τ^* on Davis, Klaeger and Anastassiadis. The ratio, the published entropy and $P_{\max}$ are the slowest scores on every dataset.

On a 50-target subpanel, a compound with one active target stays scored with probability 0.14 (Davis), 0.23 (Klaeger), 0.17 (Anastassiadis) and 0.29 (Metz), and one with ten active targets with probability 0.78, 0.93, 0.85 and 0.97. (`panel_recovery.py`)

S6: Cross-dataset and supplementary comparisons

Dataset and definition. On the 46 compounds profiled in both Davis and Klaeger and the 201 kinases both panels share, the same definition's Davis and Klaeger rankings correlate at 0.78 (S-score), 0.84 (entropy), 0.45 (Gini) and 0.34 (ratio, $p = 0.02$), and different definitions within one dataset at −0.08–1.00 (mean 0.57). Kinases are matched by gene symbol, with renamed Davis symbols mapped to current ones (for example, Davis's MLCK, accession NP_872299.2, is MYLK3). A check of the other renamed symbols found no further matches. Klaeger has six compounds with no active target on this shared panel. Among the 40 compounds active in both, the same definition's rankings correlate at 0.74, 0.85, 0.84 and 0.46 (mean 0.72, against 0.73 between definitions). On the full panels the cross-dataset correlations are 0.14–0.81 (mean 0.53).

In a separate diagnostic using the full curated panels and the original 68- and 222-compound ranking populations, aggregated rank instability is only weakly associated with cross-dataset rank displacement for these 46 compounds ($|r| \leq 0.18$). This diagnostic aggregates all four definitions and is not a result about each definition's own parameter sensitivity. Replacing it by within-definition instability gives $r = 0.327$ for the Davis S-score (nominal $p = 0.027$; Bonferroni-adjusted $p = 0.212$ over eight tests). No within-definition association survives that correction. These exploratory results do not establish that parameter sensitivity is generally uninformative about robustness across datasets. (`davis_klaeger_overlap.py`, `platform_vs_definition.py`, `claim_checks.py` section 9)

Bosc et al.'s scores and a CATDS proxy. Bosc et al.'s[6] window score (WS) and ranking score (RS) correlate strongly with the ratio on every dataset (Table S8). RS at rank $k+1$ is $R(k)$, so its agreement with the ratio follows from its definition. CATDS scores one designated target at a chosen concentration from Kinobeads curves, which only Klaeger provides. A proxy, the top target's share of the baseline-subtracted profile, correlates more with the Gini than with the ratio on every dataset, and more with the entropy than with the ratio except on Klaeger. It lacks the concentration dependence of the published score. (`bosc_catds_comparison.py`)

Table S8: Spearman correlation of WS and RS with the ratio, and of the CATDS proxy with the ratio, entropy and Gini.

Dataset	n_{cpd}	WS vs. ratio	RS vs. ratio	CATDS vs. ratio	CATDS vs. entropy / Gini
Davis	68	0.932	0.904	0.636	0.977 / 0.977
Klaeger	222	0.926	0.890	0.724	0.637 / 0.982
Anastassiadis	178	0.957	0.883	0.698	0.945 / 0.964
Metz	704	0.910	0.899	0.447	0.883 / 0.906

S7: Clinical analysis

For the 46 FDA-approved Klaeger compounds with FAERS records, we correlated each definition's median selectivity rank with the proportions of FAERS reports that were serious, fatal or involved hospitalization. FAERS counts were retrieved from the openFDA adverse-event interface (`faers_pull.py`). No correlation was statistically significant in this exploratory analysis (all $|r| \leq 0.20$, all $p > 0.19$); this does not establish the absence of clinically relevant associations. Raw report counts correlate weakly with the ratio ranking ($r = 0.30$, $p = 0.04$), which may reflect prescription volume. FAERS proportions describe reports, not incidence among treated patients,

and they depend on indication and prescription volume. The analysis therefore cannot isolate an effect of selectivity on toxicity. (`clinical_outcomes_correlation.py`)

S8: Reproducibility index

Every result in the main text is printed by a script in the code repository. Table S9 gives, for each result, the script that computes it. Each script saves its printed output as `outputs/<script>.txt`. Numbers in parentheses are numbered sections of that output. Random steps are seeded, so rerunning a script reproduces its output.

Table S9: Provenance of reported results.

Location	Result	Script (output section)
Methods; Table 1; S1	Curated panels, curation tables, Klaeger values below the floor	`curate_panels.py`
Methods; S1; Discussion	Active fractions, entries at or below the activity threshold, Anastassiadis weak-inhibition cutoff, Metz coverage before restriction, Klaeger compounds with few active kinases, Davis stricter threshold	`dataset_descriptive_stats.py`
Methods; S1	Metz coverage of the retained compounds	`metz_coverage.py`, `claim_checks.py` (10)
Methods; S2	Mean instead of median ranks	`claim_checks.py` (3)
Table 3; Results, Agreement	Correlations between definitions; Klaeger active compounds; bootstrap	`selectivity_analysis.py`, `klaeger_analysis.py`, `claim_checks.py` (2)
Table 3; Figure 1; Table S2	Published entropy and Gini, both floor treatments; every measured Klaeger affinity retained	`original_operationalization_robustness.py`, `metric_fidelity_robustness.py`
Results, Agreement; S3	Compounds with no active kinase: ranks, aggregated and per-definition instability, undefined configurations	`claim_checks.py` (13, 20), `failure_mode_illustration.py`, `undefined_score_convention.py`
Results, Agreement; S3	Targets of the compounds with no detected binding	`claim_checks.py` (1)
Results, Agreement; Figure S1; S2	Four datasets, all and active compounds; Metz coverage dependence	`additional_datasets_analysis.py`, `claim_checks.py` (10, 11)
Methods; Results, Agreement; Table S1	Released against curated panels	`davis_gene_level.py`, `klaeger_protein_kinases.py`
Results, Agreement; Discussion; S6	Dataset against definition on shared compounds and kinases (renamed symbols mapped); active fractions; instability against cross-dataset rank change	`davis_klaeger_overlap.py`, `platform_vs_definition.py`, `claim_checks.py` (9)
Results, Agreement; Results, Recovery; S1	Klaeger confidence filter, including gated p^*	`klaeger_filter_sensitivity.py`
Results, Sensitivity; Table S3; Figure S2	Pairs reversing in B; gate against baseline	`d3_pair_reversals.py`, `failure_mode_illustration.py`
Results, Sensitivity	Ratio at near-tied top targets	`d2_near_tie_sensitivity.py`
Figure 2	Binding profiles of two Davis compounds	`binding_profiles.py`

Location	Result	Script (output section)
Results, Sensitivity; Table S4; Figure S3	Profile features and instability, both ranking populations; compounds defined at every baseline; Anastassiadis and Metz relations	`instability_by_family.py`, `additional_datasets_analysis.py`, `claim_checks.py` (4, 8, 15–17, 21)
S3	Undefined features; further features	`feature_placeholder_sensitivity.py`, `instability_predictor_candidates.py`
Results, Sensitivity; S3	Broad profiles; most unstable and most stable compounds	`claim_checks.py` (5–7, 18)
Methods; Results, Criteria; Table 4; S4	Bound on T; entropy fact used in Proposition 1; D2 (locality); entropy and gated entropy under weak addition; worked example; discontinuity; strengthening example; sharpness; D4 on reference profiles and below the floor; Proposition 3	`d4_entropy_vs_candidate.py`
Results, Criteria	Realistic panel expansion	`claim_checks.py` (19)
Results, Criteria; Table S5	Correlation with the number of active kinases, D3 at τ^*, scale T	`candidate_validation.py`
Results, Criteria	Agreement with the entropy	`candidate_measure.py`
Results, Criteria; S4	Dependence on τ^*; noise	`claim_checks.py` (22, 23)
Results, Criteria; Results, Recovery; Figure 3; S5	Coverage, recoverability, conditional agreement, variability, influence, agreement with the S-score at x_0; other bars; probability of staying scored	`panel_recovery.py`
Results, Scale; S5	Gated entropy with $T = 2(\tau^* - x_0)$ and $4(\tau^* - x_0)$: recovery, agreement with the S-score at x_0	`panel_recovery.py`
Results, Criteria; S4; Table S6	Variants of the gated entropy	`family_construction.py`, `family_panelsize.py`
S5; Figure S4; Table S7	Recovery without D1	`panel_size_analysis.py`, `pstar_bar_sensitivity.py`, `claim_checks.py` (14)
Discussion; S6; Table S8	Bosc et al.'s scores and the CATDS proxy	`bosc_catds_comparison.py`
Discussion; S7	Clinical outcomes	`clinical_outcomes_correlation.py`, `faers_pull.py`

References

[1] Klaeger, S.; Heinzlmeir, S.; Wilhelm, M.; et al. The target landscape of clinical kinase drugs. *Science* **2017**, *358*, eaan4368.

[2] Gao, Y.; Davies, S. P.; Augustin, M.; Woodward, A.; Patel, U. A.; Kovelman, R.; Harvey, K. J. A broad activity screen in support of a chemogenomic map for kinase signalling research and drug discovery. *Biochem. J.* **2013**, *451*, 313–328.

[3] Patricelli, M. P.; Nomanbhoy, T. K.; Wu, J.; et al. In situ kinase profiling reveals functionally relevant properties of native kinases. *Chem. Biol.* **2011**, *18*, 699–710.

[4] Shannon, C. E. A mathematical theory of communication. *Bell Syst. Tech. J.* **1948**, *27*, 379–423.

[5] Karaman, M. W.; Herrgard, S.; Treiber, D. K.; et al. A quantitative analysis of kinase inhibitor selectivity. *Nat. Biotechnol.* **2008**, *26*, 127–132.

[6] Bosc, N.; Meyer, C.; Bonnet, P. The use of novel selectivity metrics in kinase research. *BMC Bioinformatics* **2017**, *18*, 17.